

Where to Put the Two Engines

Setting the twin-engine Apollo LM's spacing from measured authority instead of by assumption

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Contents

1	The parameter that was never justified	1
2	What the spacing does and does not change	1
3	The sweep	2
4	Three constraints, and a threshold that appears twice	2
5	Recommendation	3
6	Recommendation	4
7	Limitations	4
8	Reproducing	4

1 The parameter that was never justified

Every fault study in this repository flies an imaginary vehicle: the Apollo LM's single Descent Propulsion System engine split into two half-thrust engines mounted at $y = \pm y_{eng}$. Mass, inertia, total thrust, gimbal throw and the entire RCS are the real LM's. Only the placement is invented — and it was invented twice, both times by assumption: **1.50 m** in Studies A through G, **0.25 m** in Study H after the one-engine-out roll budget showed 1.50 m could never be trimmed.

Neither number was ever checked against what an Apollo-class lander's rotational authority actually is. The fleet actuation-envelope survey now supplies that reference, and it supplies it for the most relevant vehicle possible: **the real Apollo LM itself**. This document sets y_{eng} by measurement.

2 What the spacing does and does not change

Both vehicles are built and measured with the same machinery used for the fleet survey — the support function of the achievable force set and of the achievable moment set, evaluated on each body axis. Body axes are the flight model's: x forward, y right, z **down along the thrust line**, so rotation about x is roll, about y pitch, about z yaw.

The sweep shows immediately that only two of the six quantities are in play:

Quantity	Real LM	Twin-engine, $y_{eng} = 0 \rightarrow 2$ m	Why
Linear a_x, a_y, a_z	0.77, 0.77, 6.07 m/s ²	unchanged at every spacing	total thrust and mass are fixed; a moment arm cannot make force
Pitch α_y	148 °/s ²	unchanged at every spacing	pitch comes from gimbal deflection against dz_{eng} , which y_{eng} does not touch
Roll α_x	148 °/s ²	148 $\rightarrow$ 564 °/s ²	differential throttling between the two engines, arm y_{eng}
Yaw α_z	34 °/s ²	34 $\rightarrow$ 141 °/s ²	pitch-gimbal deflection acting through the lateral offset

So the question is narrower than it looks. Splitting the engine cannot change what the vehicle can accelerate at, and cannot change its pitch authority. What it does is **invent roll and yaw authority that the real vehicle never had** — and the further apart the engines, the more of it.

3 The sweep

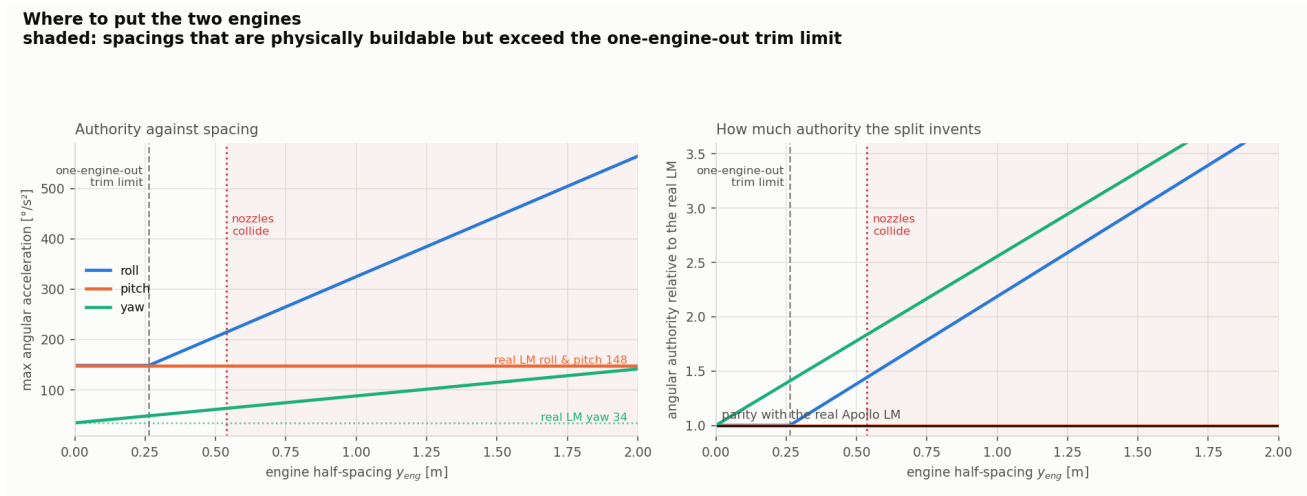


Figure 1: Angular authority against engine spacing

y_{eng} [m]	Roll [°/s ²]	Pitch [°/s ²]	Yaw [°/s ²]	Roll / LM	Pitch / LM	Yaw / LM
0.00	148	148	34	1.00	1.00	1.00
0.10	148	148	40	1.00	1.00	1.16
0.25	148	148	48	1.00	1.00	1.39
0.50	205	148	61	1.38	1.00	1.78
0.75	265	148	75	1.78	1.00	2.17
1.00	325	148	88	2.19	1.00	2.56
1.50	444	148	115	2.99	1.00	3.33
2.00	564	148	141	3.80	1.00	4.11

4 Three constraints, and a threshold that appears twice

Roll parity. Roll authority stays *exactly* at the real LM's 148 °/s² up to $y_{eng} = 0.260$ m, and climbs from there. Below that spacing the best roll moment the vehicle can make is still the one the single-engine LM makes

— both gimbals deflected in yaw against the 2.5 m engine-plane offset. Differential throttling is the better lever only once the arm is longer than that.

One-engine-out trimmability. Study A’s budget, derived independently and from entirely different reasoning: a single gimbal can trim the roll asymmetry left by a dead engine only while $y_{eng} \leq dz_{eng} \tan \delta_{max} = \mathbf{0.263\ m}$.

Those two numbers are the same, and it is not a coincidence. Both ask the same question — *is the lateral moment arm longer than the moment arm the gimbal already commands?* Below $dz_{eng} \tan \delta_{max}$ the gimbal dominates the roll axis, so the split adds nothing and a surviving gimbal can undo what a dead engine does. Above it, differential thrust dominates, the vehicle gains roll authority it never had, and the same arm that gave it that authority is what the lone gimbal can no longer overcome. **One threshold governs both the fidelity of the model and the survivability of the fault.**

Yaw cannot be held at parity at all. Unlike roll, yaw authority rises from the very first centimetre of separation — 34 °/s² on the centreline, 40 at 0.10 m, 48 at 0.25 m — because pitch-gimbal deflection acting through the lateral offset makes a yaw moment that a centreline engine simply cannot. There is no positive spacing at which yaw is at parity, so yaw cannot be a criterion: the only spacing that satisfies it is no split at all.

The reason yaw inflates so readily is that it is the LM’s *weakest* axis by a factor of four (34 °/s² against 148 for roll and pitch). Almost all of the real vehicle’s yaw authority comes from the RCS, so any engine contribution at all is a large *relative* change while staying small in absolute terms: at 0.25 m the split adds 13 °/s² to an axis that has 34.

Nozzle geometry. Two engines cannot overlap. The real DPS nozzle is about 1.52 m across; at fixed chamber pressure and expansion ratio the exit area scales with thrust, so a half-thrust engine is $1/\sqrt{2}$ as wide — 1.07 m. Two of them cannot sit closer than one diameter centre to centre, giving $y_{eng} \geq \mathbf{0.54\ m}$.

Constraint	Requires	Source
Roll authority at parity with the real LM	$\leq 0.260\ \text{m}$	measured from the sweep
One-engine-out trimmable at 6° gimbal	$\leq 0.263\ \text{m}$	Study A roll budget — the same threshold
Yaw authority at parity	$\leq 0.00\ \text{m}$	degenerate; rejected
Nozzles do not collide	$\geq 0.54\ \text{m}$	estimated from a sqrt(T) nozzle scaling

The buildable region and the trimmable region do not overlap. Anything a real pair of half-DPS engines could be built at ($\geq 0.54\ \text{m}$) is already past the threshold where one gimbal can trim the other’s failure, and past the point where the split starts inventing roll authority ($\leq 0.263\ \text{m}$).

That is a finding about the configuration rather than a modelling inconvenience: **a twin-engine Apollo-class lander with only 6° of gimbal throw cannot be both buildable and single-engine-out survivable.** The gimbal is the binding parameter, not the spacing. Making the buildable spacing trimmable needs

$$\delta_{max} \geq \arctan \frac{y_{eng}}{dz_{eng}} = \arctan \frac{0.54}{2.5} = 12.1^\circ$$

— a little over double the LM’s actual throw, which is a large but not absurd ask for a clean-sheet design.

5 Recommendation

Keep $y_{eng} = 0.25\ \text{m}$, and state it as a *modelling* choice rather than a design one.

It is the largest round spacing under the 0.263 m threshold, so the imaginary vehicle has **exactly** the real Apollo LM’s roll and pitch authority, and it is engine-out trimmable — which is what turns engine-out from an automatic loss into a fault with an answer. The price is 39 % more yaw authority than the real LM, on the axis where the real LM is weakest and where the absolute difference is 13 °/s². That is the residual, and it should be quoted rather than hidden.

What the vehicle is not is buildable. Two half-DPS engines 0.50 m apart centre to centre would overlap, so it should be described as **a device for splitting the DPS into two independently faultable halves while holding the real LM’s authority fixed** — which is exactly what the fault studies need — and not as a proposal for a real lander.

Studies A through G, at 1.50 m, should carry the caveat that their vehicle had 3.0x the roll and 3.3x the yaw authority of a real LM. Their engine-out result is unaffected — that fault was lost to the trim budget, which 1.50 m misses by a factor of 5.7 — but their gimbal-fault and thrust-asymmetry results were measured on an over-actuated vehicle.

The buildable region and the trimmable region do not overlap. Anything a real pair of half-DPS engines could be built at (≥ 0.54 m) is already past the point where one gimbal can trim the other’s failure (≤ 0.263 m), and past parity with the real LM’s authority.

That is a genuine finding about the configuration rather than a modelling inconvenience: **a twin-engine Apollo-class lander with only 6° of gimbal throw cannot be both buildable and single-engine-out survivable.** The gimbal is the binding parameter, not the spacing. Making the buildable spacing trimmable needs

$$\delta_{max} \geq \arctan \frac{y_{eng}}{dz_{eng}} = \arctan \frac{0.54}{2.5} = 12.1^\circ$$

— a little over double the LM’s actual throw, which is a large but not absurd ask for a clean-sheet design.

6 Recommendation

Keep $y_{eng} = 0.25$ m for the fault studies, and state it as a *modelling* choice rather than a design one.

At that spacing the imaginary vehicle matches the real Apollo LM’s roll authority exactly and its yaw authority to within 39 %, so the fault campaigns are flying something with an Apollo-class lander’s actual pointing capability rather than an invented one. It also sits just inside the one-engine-out trim limit, which is what makes engine-out a fault with an answer instead of an automatic loss.

What it is not is buildable. Two half-DPS engines 0.50 m apart centre to centre would overlap, so the vehicle should be described as **a device for splitting the DPS into two independently faultable halves while holding the real LM’s authority fixed** — which is exactly what the fault studies need it to be — and not as a proposal for a real lander.

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7 Limitations

- **The nozzle constraint is an estimate.** 1.52 m for the DPS exit diameter and a $\sqrt{T}$ scaling for the half-thrust engine are both approximations. A genuinely optimised half-thrust engine at a different expansion ratio could be meaningfully smaller, which would move the buildable limit down — though not to 0.263 m.
- **Authority is an upper bound.** As in the fleet survey, every actuator fires at once in the optimal direction, and angular acceleration is M/I_{jj} about principal axes with no gyroscopic coupling.
- **Inertia is held fixed** across the sweep. Moving real engines outboard would change I_{xx} slightly; at these masses and arms the effect is well below the shape-model uncertainty already present in the inertia itself.
- **Only the spacing is varied.** Engine cant, axial station and differential gimbal throw are all fixed at the LM’s values, and each is another lever on the same trade.

8 Reproducing

python studies/engine_placement/place_engines.py